

Without a Compass: How the Ancients Determined Direction

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Abstract

A Song-dynasty book, the *Pingzhou ketan*, records that the master of a seagoing ship watches the stars at night, the sun by day, and the compass only in overcast weather — the compass standing third. This note deals with the long stretch before the needle appeared, and with how people nevertheless fixed their directions. The core method, given in the *Zhouli*, "Kaogong ji", is the gnomon: level the ground, set up a pole, draw a circle, and mark on the circumference the tip of the pole's shadow at sunrise and again at sunset. The two points are left-right symmetric about the line of true north-south; joining them gives true east-west, and folding the line back to the pole gives true north-south, so that no direction need be known in advance. At night the pole star, the one place that barely moves, and the Dipper, a dial that turns, serve as rulers. The note then sets out the archaeological evidence — the Taosi observation structure in Shanxi and tomb 45 at Xishuipo in Henan — and traces the word "sinan" from a gnomon pole to a ladle-shaped object to the magnetised needle. It closes with the way orientation entered ritual and speech. The note is descriptive; it contains no predictive claims.

Keywords: direction-finding; gnomon; shadow measurement; pole star; the Dipper; sinan; magnetic declination; Taosi; Xishuipo

1. One pole, one circle, two shadows

There is a Song-dynasty book called the *Pingzhou ketan*, by Zhu Yu, which records a rule observed aboard a seagoing ship:

舟师识地理，夜则观星，昼则观日，阴晦观指南针。(The ship's master knows the land: at night he watches the stars, by day he watches the sun, and in overcast weather he watches the

compass.)

The ship's navigator knows his way in three grades. At night he watches the stars, by day the sun, and only in overcast weather does the compass come round.

The compass comes third.

The point of importance here is this: on a Song-dynasty ship the compass was a substitute. In fine weather it was not needed.

A word on chronology first. This record comes from the eleventh or twelfth century, when the magnetic needle was already aboard ship. What follows concerns the earlier and much longer stretch, before the needle appeared, when people nonetheless fixed out east, west, north and south.

Begin with a passage in the *Zhouli*, "Kaogong ji", section "Craftsmen":

匠人建国，水地以县，置槲以县，眡以景。为规，识日出之景与日入之景，昼参诸日中之景，夜考之极星，以正朝夕。（The craftsman establishes the state: he levels the ground by means of the plumb line, sets up the pole by means of the plumb line, and looks to the shadow. He draws a circle, marking the shadow at sunrise and the shadow at sunset, by day checking against the noon shadow, by night examining the pole star, and so corrects morning and evening.）

The words are few, but the unfamiliar characters are dense; each must be taken apart.

县 here is 悬, "to suspend". A suspended cord gives a straight line, and water is used to test level. "To level the ground with a plumb line" is to make the ground flat.

槲 is read niè and is interchangeable with 杲; it is the gnomon, the pole set upright in the middle of level ground.

眡 is 视, "to look". 景 is 影, "shadow", the shadow of the sun.

为规 is to draw a circle with the pole as its centre. This step is the crux; without the circle there is nothing to compare the two shadows against.

The two characters 朝夕 are the most easily misread. They are not "morning and evening". On "以正朝夕", Zheng Xuan's Han-dynasty commentary and Sun Yirang's Qing-dynasty subcommentary are both explicit: to rectify zhao-xi is to rectify east and west.

Put together: level the ground, set up a pole, and draw a circle with the pole as centre. At sunrise mark the point where the tip of the pole's shadow falls on the circumference; at sunset mark another.

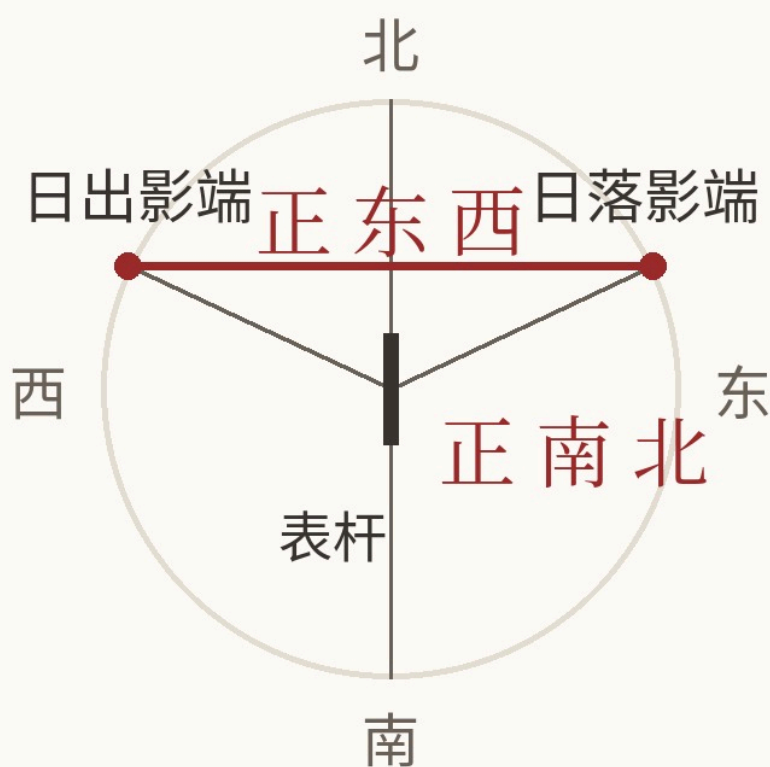
The *Zhouli* *suanjing* sets the method out more plainly:

其术曰：立正勾定之。以日始出，立表而识其晷；日入复识其晷。晷之两端相直者，正东西也；中折之指表者，正南北也。（The method says: set up a true gnomon and fix it. At the first rising of the sun, set up the gnomon and mark its shadow; at sunset, mark the shadow again. The two ends of the shadows, joined straight, give true east and west; halved at the middle and pointed at the gnomon, they give true north and south.）

The moment the sun comes up, mark the shadow once; when it goes down, mark it again. Join the two ends of the two shadows, and that line is true east-west. Fold the line at its middle, and the midpoint so obtained, joined to the pole, is true north-south.

One pole, one circle, two marks a day. Direction comes out.

立表测影 两根影，定出东西南北



同一天，日出与日落的照射角相同
两条影因此左右对称，对称轴就是正南北

2. Why two shadows suffice

The reasoning is a single sentence: on the same day, the sun's angle of illumination at sunrise and at sunset is the same.

The sun rises in the east and sets in the west; its light comes from directions that are symmetric between morning and evening. The tips of the two shadows are therefore left-right symmetric about the line of true north-south.

The axis of symmetry is true north-south.

The beauty of the matter: at no point does it require knowing any direction in advance. With a pole in hand and level ground underfoot, and two shadow marks a day, direction comes out of itself.

One thing must be explained. On the two days of the spring and autumn equinoxes the sun rises due east and sets due west; the pole's shadow and the pole fall almost on the same east-west line, and the joining of the two ends does not hold. The *Zhoubi suanjing* also leaves a line: 冬至日加卯酉者……日不见矣, "on the day of the winter solstice at the hours mao and you ... the sun is not seen"; there are days on which this cannot be measured at all.

So this must be done on chosen days, and repeated several times. It is not settled by a single measurement.

3. By day the sun, by night the stars

The second half of that passage in the *Kaogong ji* is 夜考之极星, "at night examine the pole star". When the day's work is done, the pole star is used at night to check it over.

The *Huainanzi*, "Qi su xun", has a line, excellent:

夫乘舟而惑者，不知东西，见斗极则寤矣。（One who is lost aboard a boat, not knowing east from west, on seeing the Dipper and the pole star at once understands.）

A person lost in direction aboard a boat, unable to tell east from west, at once understands on seeing the Dipper and the pole star.

Why can the pole star serve as true north? Because the earth's axis points at it. All the stars in the sky turn through the night; only the pole star barely moves. To pick out one "unmoving" thing from a whole sky of "moving" ones is to obtain a fixed mark of direction.

One reminder here. The "pole star" spoken of today is today's star. The celestial pole itself is also slowly moving, and the star nearest the pole two or three thousand years ago was not

today's. Hence the old books write "pole star" or "northern pivot" (北辰), which is a position, not the fixed name of a star. Behind this lies precession, which has a note of its own.

The Dipper, for its part, is the most serviceable ruler at night. The *Shiji*, "Treatise on the Celestial Offices", says:

斗为帝车，运于中央，临制四乡。分阴阳，建四时，均五行，移节度，定诸纪，皆系于斗。(The Dipper is the emperor's chariot, revolving at the centre and governing the four quarters. The dividing of yin and yang, the establishing of the four seasons, the balancing of the five phases, the shifting of the measured degrees, the fixing of the several registers — all depend on the Dipper.)

The Dipper is the emperor's chariot, revolving at the centre and governing the four quarters. Where the "centre" in the sky is, the four directions are thereby divided out.

The *Yizhou shu*, "Zhou yue jie", also leaves a plainly written record: 斗柄建子，始昏北指, "the handle of the Dipper establishes 子; at dusk it points north". At dusk the handle of the Dipper points north.

It must be made clear, though: the Dipper is not nailed in place pointing at one direction. The earth rotates, and it circles the celestial pole all night, so it is a dial that turns, and one must know when to look at it. The Dipper can also be used to fix the month, which is another matter, treated earlier.

4. Orientation was not reasoned out; it was dug up

What has been said above is all words on paper. Is there anything solid?

There is. And it is far earlier than paper.

At the Taosi site in Xiangfen, Linfen, Shanxi, the sacrificial area contains a structure numbered II FJT1, roughly semicircular and covering about one thousand seven hundred and forty square metres. It takes the form of thirteen rammed-earth pillars enclosing twelve observation slots; the pillars stand at a radius of about ten point five metres from the centre, and the slots are mostly between fifteen and fifty centimetres wide, a width that roughly corresponds to the apparent diameter of the sun to the eye.

The use is this: a person stands at the fixed observation point at the centre and looks east, to see through which slot the sun passes as it rises over Mount Ta'er opposite. The slot is fixed; the sun moves. This reading is the excavators' view, and voices of reservation are still heard in the field.

The structure as a whole faces south-east. Its method is precisely to use a fixed reference point to measure the direction of sunrise. From a large tomb of the middle Taosi period a painted wooden gnomon rule with graduations was also unearthed, the earliest physical example of a gnomon instrument so far found anywhere by archaeology.

He Nu, a researcher at the Institute of Archaeology of the Chinese Academy of Social Sciences, summed the whole thing up in four characters: 经天纬地, "to lay out the sky and weave the earth". To lay out the sky is to fix the calendar and reverently to bestow the farming seasons; to weave the earth is to distinguish the directions and set the positions, to establish the centre and set up the pole.

The two characters "weave the earth" speak precisely of fixing direction.

An earlier site is at Puyang in Henan. Tomb 45 of the Xishuipo site, excavated in 1987, belongs to the Yangshao culture period, about six thousand five hundred years ago.

Researchers noticed one thing: four groups of remains — tomb 45 itself plus three subsidiary shell-mosaic features — are laid out strictly on the same meridian, evenly spaced, one group every twenty-five metres southward. On this basis it is judged that such an arrangement is no accident, and that the people of the time could already determine the meridian and used it to lay out the whole space.

The mosaics in the tomb are these: on the east side of the tomb occupant a dragon in shell, on the west side a tiger in shell; at the feet a triangular figure in shell, read as the Dipper, its handle laid out with two human shinbones. Researchers hold that two methods — setting up the gnomon to measure shadows, and observing the Dipper — are placed here together.

Taken together, these two sites show one thing: orientation was not an idea slowly groped toward. Six thousand five hundred years ago it was already set out precisely on the ground.

5. From a pole to a needle

Return to that order at the beginning. Before the compass, people relied on the pole, the shadow and the stars.

Where, then, do the two characters 司南 (sinan) come from?

Many do not know that it was not originally a magnetic spoon. The Han Feizi, "You du", has a line:

故先王立司南，以端朝夕。(Therefore the former kings set up the sinan, to rectify morning and evening.)

To rectify zhao-xi is to rectify east and west, the same sense as 以正朝夕 in the Kaogong ji. So the earliest "sinan" means a gnomon pole for measuring shadows.

The Guiguzi, "Mou pian", says:

郑人之取玉也，必载司南之车，为其不惑也。（When the men of Zheng went to fetch jade, they would carry a sinan cart, so as not to lose their way.）

When the men of Zheng went to gather jade, they carried a sinan on the cart, so as not to lose the way. By this point the sinan could be carried along.

By the Eastern Han, Wang Chong wrote in the Lunheng, "Shi ying pian":

司南之杓，投之于地，其柢指南。（The ladle of the sinan, set down on the ground, has its handle point south.）

By then the sinan was a ladle-shaped object which, set on the ground, has its handle pointing south. The character 杓 in this line is written 勺 in some editions.

After that comes magnetism. Shen Kuo of the Northern Song records an entry in the Mengxi bitan, chapter 24:

方家以磁石磨针锋，则能指南，然常微偏东，不全南也。（The masters of methods rub the point of a needle with lodestone, and it can then point south; but it usually inclines slightly east, and does not point due south.）

Rub the tip of a needle with lodestone and the needle can point south, but it usually inclines slightly east, and does not point due south. Shen Kuo set down the three characters "inclines slightly east", about four hundred years before Europeans discovered magnetic declination.

A single line now becomes clear:

Period	What direction was fixed by
Pre-Qin	Setting up the gnomon to measure shadows — one pole, two shadows
Warring States to Han	"Sinan" passes from gnomon pole to the name of an object
Northern Song	Artificially magnetised steel needles; Shen Kuo records magnetic declination
Song-Yuan and after	The magnetic needle goes aboard ship, fitted into the compass

The name changed, the instrument changed; the thing to be done was exactly the same — to fix direction.

6. Once orientation was fixed, it became a rule

Once direction was measured out, it did not stop at the level of measurement. It entered ritual, and it entered speech.

The *Zhouyi*, "Shuo gua": 圣人南面而听天下，向明而治 ("The sage faces south to hear the affairs of the world, and governs facing the light."). To face south is the sovereign's position. In the *Zhuangzi*, "Dao zhi", it becomes a set phrase altogether: 南面称孤, "facing south and calling oneself the lone one".

The reverse is to face north. The *Shiji*, "Biographies of Li Yiji and Lu Jia": 君王宜郊迎，北面称臣 ("Your Majesty should go out to the outskirts to meet him, and face north to call yourself his subject."). To face north is the subject's position.

Carriages have their own rules. The *Shiji*, "Biography of the Lord of Wei": in going to meet Master Hou, 公子从车骑，虚左 ("the prince brought the carriage and outriders and left the left-hand seat empty"), leaving the left-hand seat vacant. To leave the left vacant is the highest respect. Hence the idiom 虚左以待, "to leave the left seat vacant in expectation".

There is also the "host of the eastern road" (东道主, *dongdaozhu*). In the *Zuozhuan*, thirtieth year of Duke Xi, the state of Zheng lay east of Qin and was willing to entertain guests coming from the east, and so was called the host of the eastern road.

One point easily misunderstood must be made clear. "In the hall, facing south is the place of honour; indoors, facing east is the place of honour"; these are two different settings — deliberation in the hall and drinking indoors — and the rules differ. As for whether "left" or "right" ranks higher, the dynasties are not consistent.

A last word: 败北 (*baibei*, "to suffer defeat").

The word has nothing to do with direction. 北 is an ancient form of 背, "back". The *Shuowen jiezi*: 北，𠂔也，从二人相背 ("北 is 𠂔; it is formed from two people turning their backs to each other."). When two armies fight, the defeated turn and run with their backs to the enemy, and so it is called *baibei*. Even if the beaten army runs all the way south, it is still called *baibei*.

Closing

Return to that pole.

The passage in the *Kaogong ji* does three things: level the ground, set up a pole, draw a circle. Then wait for sunrise, wait for sunset, and mark the shadow twice.

Once in the morning, once in the evening; join them into a line and it is east-west; fold it back at the middle to point at the pole, and it is north-south.

The Huainanzi says 见斗极则寤矣, "on seeing the Dipper and the pole star, one understands"; a lost person, seeing the Dipper and the pole star, at once understands. And before ever seeing the stars, people had already measured direction out with a pole.

East, west, north and south were not written on the ground to begin with. They were measured out by people with a pole, marking the shadow twice a day.

A&Q

Without a compass, how did the ancients fix east, west, north and south?

Set up a pole, draw a circle, mark the two shadow-tips at sunrise and at sunset. Join the two ends for true east-west; fold back from the middle toward the pole for true north-south. This is 以正朝夕 in the Zhouli, "Kaogong ji".

Why are two shadow markings enough?

On the same day the sun's angle of illumination at sunrise and at sunset is the same, and the two shadow-tips are left-right symmetric about the true north-south line. So no direction need be known in advance.

Was the "sinan" originally a magnetic spoon?

No. The Han Feizi, "You du", says 立司南, 以端朝夕; to rectify zhao-xi is to rectify east and west, and it means a gnomon pole for measuring shadows. Only with Wang Chong's Lunheng, "Shi ying pian", of the Eastern Han does 司南之杓 appear, a ladle-shaped object.

Who first recorded magnetic declination?

Shen Kuo. In the Mengxi bitan, chapter 24, he writes 常微偏东, 不全南也, about four hundred years earlier than Europeans.

Does 败北 mean "to run north"?

No. 北 is an ancient form of 背, and the Shuowen jiezi says 从二人相背. A person defeated in battle runs with his back to the enemy, and so it is called baibei; whichever direction he runs, it is still called baibei.

Note by Chujia Kuang

One pole, one circle, two shadows a day: it looks crude, but the reasoning is deep. It does not require you first to know east, west, north and south; rather, it hands them to you. What is called a "sense of direction" today mostly means finding one's way; direction as the ancients spoke of it was measured accurately first, and then used to fix ritual, position and cities. Once

this level is established, later treatment of the earthly branches matched to directions, and of the three unions and three meetings, has a root.

About the author

Chujia Kuang (邝楚嘉), pen name Jiayan Yide (嘉言一得), is an independent researcher in Jieyang, Guangdong, devoted to archiving verifiable primary sources and calendar data.

This column writes about calendrics, solar terms, the stems and branches and orientation, and also about what he reads. The pen name 嘉言一得 comes from the 史记: 愚者千虑, 必有一得, "the foolish man's thousand thoughts must yield one gain". Where something is written wrongly, corrections are welcome.

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